The Badger Habitable Zone

Reconsidering the Habitability of Exoplanets Without a Bias towards *Homo Sapiens*

By

Bucky Badger

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The dissertation is approved by the following members of the Final Oral Committee:

Eleanor Burrowsworth, Professor, Astronomy
Marcus Underroot, Professor, Astronomy
Harold Earthsnout, Professor, Mammalian Studies
Lydia Snifflewhisk, Professor, Soil Engineering
Gregory P. Diggs, Research Scientist, Burrowing Institute

Dedicated to my father,
for teaching me how to dig deep.

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Under the supervision of Professor Eleanor Burrowsworth
At the University of Wisconsin-Madison

Abstract

This thesis boldly challenges the long-standing assumption that the universe exists primarily to keep Homo sapiens comfortable, hydrated, and mildly entertained. Instead, it proposes the Badger Habitable Zone (BHZ): a revolutionary framework that asks the scientific community to consider the needs of badgers—creatures who neither requested human dominance nor signed off on their planetary real-estate criteria. By analyzing exoplanet climates, soil compositions, and the availability of things badgers might enjoy digging into, the BHZ dramatically expands the list of potentially habitable worlds to include planets previously dismissed as too cold, too dark, or entirely made of mud. The findings suggest that many exoplanets are perfectly suited for life, provided that life is short, stout, and aggressively committed to burrowing. Ultimately, this work argues that current habitability metrics are embarrassingly human-centric and that the cosmos may be teeming with species who would find Earth, frankly, a bit too tidy.

Acknowledgments

I extend my deepest thanks to my family—particularly my mother, who first encouraged my early academic pursuits by proudly displaying every tunnel I dug as "evidence of intellectual promise." My father contributed invaluable support by teaching me the ancient art of strategic burrowing, a skill that proved essential both for fieldwork and for hiding from committee emails. My siblings also deserve recognition for their tireless efforts in distracting predators while I collected soil samples, even if they occasionally mistook my research notes for snacks.

I am profoundly grateful to my thesis advisor, Professor Burrowsworth, whose patience, guidance, and ability to interpret my dirt-smudged drafts made this work possible. His insistence that I "think beyond the sett" pushed me to explore exoplanetary environments far beyond the comfort of loamy soil. I also thank him for never questioning why half of my data sets smelled faintly of earthworms.

Finally, I acknowledge the broader scientific community for beginning to recognize that habitability is not solely a human affair. In particular, I would like to thank Microsoft Copilot for generating the text of this thesis. May this thesis inspire future researchers—badger, human, or otherwise—to reconsider the cosmos with a more inclusive lens.

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Introduction

Ellen: So, what do you know about cosmic rays?

Roark: They are charged particles, traveling really fast. That's about it.

— (2020), over Zoom

Chapter 2

Conclusion

I leave Sisyphus at the foot of the mountain!

One always finds one's burden again.

— *The Myth of Sisyphus* (1955) by Albert Camus